Chapter-by-Chapter Exercise Workbook

Chapter-by-Chapter Exercise Workbook

The Science of Learning SQL: An Evidence-Based, Step-by-Step Guide to Mastering Databases, Queries, Joins, and Real-World Data Analysis for Complete Beginners

---

How to Use This Workbook

Each exercise is framed as a real business question a data analyst might receive. Write your query in the blank space provided, then compare your results against the expected output table. Use this workbook for active recall — attempt every query from memory before checking the solution.

Grading key:

Mark	Meaning
✅	Output matches exactly
🟡	Correct logic, minor formatting difference
❌	Reread the relevant section and retry

Chapter 3 — Your First SELECT Statement

Business Context

You are a junior analyst at a retail company. Your manager needs readable summaries from the products table.

Exercise 3.1 (Difficulty: Beginner)

> *"List only the product name and price for every product we carry."*

```
-- Write your query here:
```

Expected Output (sample rows):

product_name	price
Wireless Headphones	49.99
USB-C Cable	12.99

Laptop Stand	34.50

Exercise 3.2 (Difficulty: Beginner)

> *"Show every unique category we sell — no duplicates."*

```
-- Write your query here:
```

Expected Output: A single-column result listing each category once (e.g., Electronics, Accessories, Furniture).

Exercise 3.3 (Difficulty: Beginner+)

> *"Display product name and price, but label the columns 'Item' and 'Retail Price' so the report looks professional."*

```
-- Write your query here:
```

- Use AS to alias both columns.
- The underlying table is unchanged after running this query.

Chapter 4 — Filtering Rows With WHERE

Business Context

The sales team needs targeted lists from the orders table.

Exercise 4.1 (Difficulty: Beginner)

> *"Find all orders where the total amount is greater than $200."*

```
-- Write your query here:
```

Expected Output (sample rows):

order_id	customer_id	total_amount
1042	305	215.00
1089	412	349.99

Exercise 4.2 (Difficulty: Intermediate)

> *"Find orders placed in the Electronics category AND with a status of 'Shipped'."*

```
-- Write your query here:
```

- Combine conditions using AND.
- Remember: text values require single quotes.

Exercise 4.3 (Difficulty: Intermediate)

> *"Identify any orders where the shipping address is missing."*

```
-- Write your query here:
```

Key reminder: Use IS NULL, not = NULL.

Chapter 5 — Sorting and Limiting Results

Business Context

Your manager wants ranked and paginated reports from the customers table.

Exercise 5.1 (Difficulty: Beginner)

> *"List the top 5 customers by total lifetime spend, highest first."*

```
-- Write your query here:
```

Expected Output (sample rows):

customer_name	**lifetime_spend**
Sarah Okonkwo	1,240.00
James Patel	980.50

Exercise 5.2 (Difficulty: Intermediate)

> *"Show customers sorted by region (A–Z), then by lifetime spend (highest first) within each region."*

```
-- Write your query here:
```

- Use two columns in ORDER BY.
- First sort column controls primary sequence; second resolves ties.

Chapter 6 — Aggregate Functions and GROUP BY

Business Context

Finance needs summary metrics from the sales table.

Exercise 6.1 (Difficulty: Intermediate)

> *"How many orders were placed in each product category?"*

```
-- Write your query here:
```

Expected Output:

category	order_count
Electronics	412
Accessories	289
Furniture	134

Exercise 6.2 (Difficulty: Intermediate+)

> *"Show only the categories where total revenue exceeds $10,000."*

```
-- Write your query here:
```

- Filter aggregated results with HAVING, not WHERE.

Chapter 7 — Joins

Business Context

You need to combine the orders and customers tables for a full sales report.

Exercise 7.1 (Difficulty: Intermediate)

> *"List every order alongside the customer's name — only where a match exists."*

```
-- Write your query here:
```

- Use INNER JOIN on customer_id.

Exercise 7.2 (Difficulty: Intermediate+)

> *"Show all customers, including those who have never placed an order."*

```
-- Write your query here:
```

- Use LEFT JOIN; unmatched order columns will appear as NULL.

Chapter 8 — Subqueries

Business Context

The analytics team needs dynamic, self-updating filters.

Exercise 8.1 (Difficulty: Advanced)

> *"Find all products priced above the average product price — without hardcoding the average."*

```
-- Write your query here:
```

- The inner query calculates AVG(price); the outer query filters against it.

Exercise 8.2 (Difficulty: Advanced)

> *"List customers who have never placed an order, using a subquery."*

```
-- Write your query here:
```

- Use NOT IN with a subquery selecting customer_id from orders.

Progress Tracker

Chapter	Exercises Attempted	✅ Correct	🟡 Partial	❌ Retry
Ch. 3 — SELECT	3			
Ch. 4 — WHERE	3			
Ch. 5 — ORDER BY / LIMIT	2			
Ch. 6 — Aggregates	2			
Ch. 7 — Joins	2			
Ch. 8 — Subqueries	2			

Tip: Any exercise marked ❌ is a retrieval opportunity, not a failure. Return to the relevant chapter section, reread the concept, then reattempt the query without looking at your previous attempt.